

$$\textcircled{1} \Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$$

$$\text{Union } \bigcup_{i \in \mathbb{N}} \Omega_i = \Omega_0 \cup \Omega_1 \cup \dots \cup \Omega_n$$

$$\text{Intersection } \bigcap_{i \in \mathbb{N}} \Omega_i = \Omega_0 \cap \Omega_1 \cap \dots \cap \Omega_n$$

$$\text{Complement } A^c = \Omega \setminus A$$

$$\text{power set } 2^\Omega = \mathcal{P}(\Omega) = \{A \mid A \subseteq \Omega\} \text{ (All subsets)}$$

$$\textcircled{2} \text{ Set : } (A \cap B)^c = A^c \cup B^c \quad \text{generalized: } \left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c$$

$$(A \cup B)^c = A^c \cap B^c \quad \text{analogously: } \left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c$$

$$\textcircled{3} \text{ samples : } \Omega = \{a, b, c\}$$

$$\text{events : } A_1 = \emptyset \quad A_2 = \{a\} \quad \{b\} \quad \{c\} \quad \{a, b\} \quad \{a, c\} \quad \{b, c\} \quad \{a, b, c\}$$

$$\text{event system : subset of } \{\emptyset, \{a\}, \{b\}, \dots, \{a, b, c\}\} \Rightarrow \mathcal{P}(\mathcal{P}(\Omega))$$

$$\textcircled{4} \text{ } \sigma\text{-Algebra : } \begin{cases} \emptyset \in \mathcal{A} \\ A \in \mathcal{A}, \text{ then } A^c \in \mathcal{A} \\ \text{for each } (A_i)_{i \in \mathbb{N}} \text{ in } \mathcal{A}, \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}. \end{cases} \quad \mathcal{A}_1 = \{\emptyset, \{a\}, \{b, c\}, \{a, b, c\}\}$$

$$\textcircled{5} \text{ Intersection } \begin{aligned} \mathcal{A}_1 &= \{\emptyset, \{a\}, \{b, c\}, \{a, b, c\}\} \\ \mathcal{A}_2 &= \{\emptyset, \{b\}, \{a, c\}, \{a, b, c\}\} \end{aligned} \Rightarrow \mathcal{A}' = \{\emptyset, \{a, b, c\}\}$$

$$\textcircled{6} \text{ For } \{a\} \quad \mathcal{L}(\{a\}) = \text{smallest } \sigma\text{-Algebra includes } \{a\}.$$

$$(\{a\} \text{ is generator of } \mathcal{L}(\{a\}))$$

Ex 1.

$$\textcircled{1} \subseteq : x \in \left(\bigcap_{\alpha} A_{\alpha}\right)^c, \exists \alpha \text{ such that } x \notin A_{\alpha}, \text{ so that } x \in A_{\alpha}^c \subseteq \bigcup_{\alpha} A_{\alpha}^c.$$

$$x \text{ is arbitrary } \Rightarrow \square.$$

$$\supseteq : x \in \bigcup_{\alpha} A_{\alpha}^c, \exists \alpha \text{ such that } x \in A_{\alpha}^c, \text{ so that } x \notin A_{\alpha}.$$

$$\text{so } x \notin \bigcap_{\alpha} A_{\alpha}, x \in \left(\bigcap_{\alpha} A_{\alpha}\right)^c. \quad \square.$$

$$\textcircled{2} \subseteq : x \in \left(\bigcup_{\alpha} A_{\alpha}\right)^c, \text{ for } \forall \alpha, x \notin A_{\alpha}, \text{ so that } x \in A_{\alpha}^c \text{ for all } \alpha.$$

$$\Rightarrow x \in \bigcap_{\alpha} A_{\alpha}^c \quad \square.$$

$$\supseteq : x \in \bigcap_{\alpha} A_{\alpha}^c, \text{ for } \forall \alpha, x \in A_{\alpha}^c, x \notin A_{\alpha} \Rightarrow x \in \left(\bigcup_{\alpha} A_{\alpha}\right)^c \quad \square.$$

Ex 2.

(i) $w \in f^{-1}(A)$, $\exists a \in A$ such that $f(w) = a \in B$
 thus $w \in f^{-1}(B)$, w arbitrary $\Rightarrow f^{-1}(A) \subset f^{-1}(B)$.

(ii) $w \in f^{-1}\left(\bigcup_{A \in \mathcal{C}} A\right)$, $\exists A \in \mathcal{C}, a \in A$ with $f(w) = a$.

$\Rightarrow w \in f^{-1}(A) \subseteq \bigcup_{A \in \mathcal{C}} f^{-1}(A)$. $\square$.

$w \in \bigcup_{A \in \mathcal{C}} f^{-1}(A)$, $\exists A \in \mathcal{C}, a \in A$ with $f(w) = a$.

Since $a \in \bigcup_{A \in \mathcal{C}} A$, $\Rightarrow w \in f^{-1}\left(\bigcup_{A \in \mathcal{C}} A\right)$. $\square$.

(iii) $w \in f^{-1}(A^c)$, $\exists a \in A^c$ such that $f(w) = a \in A^c$

$\Rightarrow w \notin f^{-1}(A)$, $w \in (f^{-1}(A))^c$

$w \in (f^{-1}(A))^c$, $\exists a \in A^c$ such that $f(w) = a \in A^c$

$\Rightarrow w \in f^{-1}(A^c)$.

Ex 3.

① (property ii, $A \in \mathcal{A}$ then $A^c \in \mathcal{A}$), $\Omega \setminus \emptyset = \Omega \in \mathcal{A}$.

② Complement of A_i must be in $\mathcal{A}$. (Follow to De Morgan $A \cap B = (A^c \cup B^c)^c \in \mathcal{A}$)
 $A_0 \setminus A_i = A_0 \cap A_i^c \in \mathcal{A}$.

③ $B_i = \begin{cases} A_i & i \leq m \\ \emptyset & i > m \end{cases}$ A_i with $i > m$ replace by $\emptyset$
 $A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i = \bigcup_{i \in \mathbb{N}} B_i \in \mathcal{A}$ (Property 3, union of ... should be ...)

④ with ①: $A_0^c, A_1^c, \dots, A_m^c \in \mathcal{A}$

with ③: $\bigcup_{i=0}^m A_i^c \in \mathcal{A}$

De-Morgan: $\left(\bigcap_{i=0}^m A_i\right)^c = \bigcup_{i=0}^m A_i^c \Rightarrow \bigcap_{i=0}^m A_i = \left(\bigcup_{i=0}^m A_i^c\right)^c \in \mathcal{A}$

⑤ $\bigcup_{i \in \mathbb{N}} A_i^c \in \mathcal{A}$ (property)

$\bigcap_{i \in \mathbb{N}} A_i = \left(\bigcup_{i \in \mathbb{N}} A_i^c\right)^c \in \mathcal{A}$.

Ex 4. (i) F. (ii) F. e.g. $A_1 = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}$
 $A_2 = \{\emptyset, \{1, 3\}, \{2, 4\}, \Omega\}$

(iii) (a) $\emptyset$ countable, $\emptyset \in \mathcal{A}$

(b) Let $A \in \mathcal{A}$, then proof $A^c \in \mathcal{A}$

co-countable: a set whose complement is countable.

- { ① A countable, then the complement of $A^c = A \in \mathcal{A}$, A^c should be in it
 ② A^c countable, the complement of $A \in \mathcal{A}$.

(c) To proof: $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$, countable or co-countable

① All A_i countable $\checkmark$

② $\exists j \in \mathbb{N}$, such that A_j is uncountable and only A_j^c countable

$$\Rightarrow (\bigcup_{i \in \mathbb{N}} A_i)^c = \bigcap_{i \in \mathbb{N}} A_i^c \subseteq A_j^c, \text{ so } \bigcup_{i \in \mathbb{N}} A_i \text{ co-countable}$$

Ex 5. ① Every $\mathcal{A}_\alpha$ includes Ω , so $\Omega \in \mathcal{A}$.

② If $A \in \mathcal{A}$, then $A \in \mathcal{A}_\alpha$ for $\forall \alpha$
 ($\mathcal{A}_\alpha$ are δ -Algebra) $\Rightarrow A^c \in \mathcal{A}_\alpha$ for $\forall \alpha \Rightarrow A^c \in \mathcal{A}$.

③ Let $A_1, A_2 \dots$ in $\mathcal{A}$
 then $A_1, A_2 \dots$ in $\mathcal{A}_\alpha$ for each α .

($\mathcal{A}_\alpha$ are δ -Algebra) $\Rightarrow A_1 \cup A_2 \dots = \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}_\alpha \Rightarrow \in \mathcal{A}$.

Ex 6.

① (a) $\Omega_2 \in \mathcal{B}$, $f^{-1}(\Omega_2) = \Omega_1 \in f^{-1}(\mathcal{B})$.

② Try yourself!

(b) Let $A \in f^{-1}(\mathcal{B})$, then $\exists B \in \mathcal{B}$, $A = f^{-1}(B)$.

($\mathcal{B}$ is δ -...) $\Rightarrow B^c \in \mathcal{B}$, $f^{-1}(B^c) = (f^{-1}(B))^c = A^c$.
 (Ex 2 iii)

$\Rightarrow A^c \in f^{-1}(\mathcal{B})$.

(c) Let $A_1, A_2 \dots \in f^{-1}(\mathcal{B})$, $\exists B_i \in \mathcal{B}$, $A_i = f^{-1}(B_i)$.

($\mathcal{B}$ is δ -...) $\Rightarrow \bigcup_{i \in \mathbb{N}} B_i \in \mathcal{B}$, $f^{-1}(\bigcup_{i \in \mathbb{N}} B_i) = \bigcup_{i \in \mathbb{N}} f^{-1}(B_i) = \bigcup_{i \in \mathbb{N}} A_i$
 (Ex 2 ii)

$\Rightarrow \bigcup A_i \in f^{-1}(\mathcal{B})$.